

Assignment on Module 03: Subnetting & VLSM

Course: Cyber Security (Cyber 13)

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4. **References:** “Cisco CCNA Routing and Switching Study Guide”

Task 1 — Basic Subnetting

Given:

- Network: 192.168.10.0/24
- Dividing this network into 4 equal subnets.
- 1. To make 4 subnets, we need 2 extra bits because $2^2 = 4$. So, we borrow 2 bits from the host part.
- 2. Adding these 2 bits to /24(255.255.255.0) gives a new prefix of /26 (Subnet Mask: 255.255.255.192). **Total IPs in each subnet:** $2^6 = 64$ total IPs.
- 3. **Usable hosts:** $2^6 - 2 = 62$ usable hosts (64 - 2 = 62, since the first IP is the network address and the last IP is the broadcast address).
- 4. **Block size:** 256 - 192 = 64 (each subnet increases by 64).

- 5. **Subnet Details:**
 - **Subnet 1:**
 - Network Address: 192.168.10.0
 - Subnet Mask: 255.255.255.192 (or /26)
 - First Usable IP: 192.168.10.1
 - Last Usable IP: 192.168.10.62
 - Broadcast Address: 192.168.10.63
 - Usable Hosts: 62
 - **Subnet 2:**
 - Network Address: 192.168.10.64
 - Subnet Mask: 255.255.255.192 (or /26)
 - First Usable IP: 192.168.10.65
 - Last Usable IP: 192.168.10.126
 - Broadcast Address: 192.168.10.127
 - Usable Hosts: 62
 - **Subnet 3:**
 - Network Address: 192.168.10.128
 - Subnet Mask: 255.255.255.192 (or /26)
 - First Usable IP: 192.168.10.129
 - Last Usable IP: 192.168.10.190
 - Broadcast Address: 192.168.10.191
 - Usable Hosts: 62
 - **Subnet 4:**
 - Network Address: 192.168.10.192
 - Subnet Mask: 255.255.255.192 (or /26)
 - First Usable IP: 192.168.10.193
 - Last Usable IP: 192.168.10.254
 - Broadcast Address: 192.168.10.255
 - Usable Hosts: 62

Task 2 — Subnetting with Host Requirements

Given:

- Network: 172.16.0.0/24
 - Requirement: At least 50 usable hosts per subnet.
1. For 50 usable hosts, we need at $(2^6 = 64)$.
 2. Subnet mask prefix: $32 - 6 = /26$ (255.255.255.192).
 3. Total subnets created: $2^{26-24} = 4$ subnets.
 4. Block size: $256 - 192 = 64$.

Subnet Details:

- **Subnet 1:**
 - Network Address: 172.16.0.0
 - Subnet Mask: 255.255.255.192 (or /26)
 - First Usable IP: 172.16.0.1
 - Last Usable IP: 172.16.0.62
 - Broadcast Address: 172.16.0.63
 - Usable Hosts: 62
- **Subnet 2:**
 - Network Address: 172.16.0.64
 - Subnet Mask: 255.255.255.192 (or /26)
 - First Usable IP: 172.16.0.65
 - Last Usable IP: 172.16.0.126
 - Broadcast Address: 172.16.0.127
 - Usable Hosts: 62
- **Subnet 3:**
 - Network Address: 172.16.0.128
 - Subnet Mask: 255.255.255.192 (or /26)
 - First Usable IP: 172.16.0.129
 - Last Usable IP: 172.16.0.190
 - Broadcast Address: 172.16.0.191
 - Usable Hosts: 62
- **Subnet 4:**
 - Network Address: 172.16.0.192
 - Subnet Mask: 255.255.255.192 (or /26)
 - First Usable IP: 172.16.0.193
 - Last Usable IP: 172.16.0.254
 - Broadcast Address: 172.16.0.255
 - Usable Hosts: 62

Task 3 — VLSM Challenge

Given:

- Network: 192.168.50.0/24
- Requirements:
 - Department A: 100 hosts
 - Department B: 50 hosts
 - Department C: 25 hosts
 - Department D: 10 hosts

VLSM Scheme & Details:

- **1. Department A (100 hosts):**
 - Host needed: 100, so we need $2^7 = 128$.
 - Network Address: 192.168.50.0
 - CIDR Prefix: /25
 - Subnet Mask: 255.255.255.128
 - First Usable IP: 192.168.50.1
 - Last Usable IP: 192.168.50.126
 - Broadcast Address: 192.168.50.127
 - Usable Hosts: 126
- **2. Department B (50 hosts):**
 - Host needed: Host needed: 50, so we need $2^6 = 64$.
 - Network Address: 192.168.50.128
 - CIDR Prefix: /26
 - Subnet Mask: 255.255.255.192
 - First Usable IP: 192.168.50.129
 - Last Usable IP: 192.168.50.190
 - Broadcast Address: 192.168.50.191
 - Usable Hosts: 62
- **3. Department C (25 hosts):**
 - Host needed: 25, so we need $2^5 = 32$.
 - Network Address: 192.168.50.192
 - CIDR Prefix: /27
 - Subnet Mask: 255.255.255.224
 - First Usable IP: 192.168.50.193
 - Last Usable IP: 192.168.50.222
 - Broadcast Address: 192.168.50.223
 - Usable Hosts: 30
- **4. Department D (10 hosts):**
 - Host needed: 10, so we need $2^4 = 16$.
 - Network Address: 192.168.50.224
 - CIDR Prefix: /28
 - Subnet Mask: 255.255.255.240
 - First Usable IP: 192.168.50.225

- Last Usable IP: 192.168.50.238
- Broadcast Address: 192.168.50.239
- Usable Hosts: 14

Reference:

Book: *Cisco CCNA Routing and Switching Study Guide* by Todd Lammle.

Website: Cisco Networking Academy (NetAcad) learning materials.